

What was inspected

97 natural eval-val proteins, exp277 epoch 2, step 479417. Original atlas: 100 saved rollout-and-resample predictions per protein. Follow-up: 200 fresh whole-map samples per protein; all 19,400 completed. All 1,940 original metric rows reproduced exactly. HTML atlas shows every predicted/true map, votes and errors, with zoom. No eval-test analysis.

The strongest visual pattern

Severe failures replace long-range traces with near-diagonal contact clusters. Worst accuracy quartile: 69.3% long-range truth vs 41.2% predictions. Examples: 8arl_A, 7wz5_A, 8bux_A, 8dmu_A. Better maps recover broad features but often miss detailed contacts. Positive control: 7y8h_A reaches 84.3% exact precision.

Why visual resemblance can mislead

Exact precision: 55.75%. Many-to-one proximity within two residues: 80.72%. Keep exact matches and pair FP with FN one-to-one: only 64.01%. The truth-informed local correction gains 8.26 percentage points. Most errors remain; local proximity is not a substitute for exact accuracy.

Sampling support in the original atlas

93.9% of true contacts appear in at least one rollout. 29.3% get fewer than ten votes; top-L/5 precision is 81.2%. Correct contacts are often present but weakly ranked. Marginal votes cannot reveal within-rollout consistency or competing folds. The follow-up below tests individual sample quality and selection. Training causes remain hypotheses; these maps do not establish one.

Whole-map probe: selection does not improve the mean

Pooled top-R: 55.79% F1; mean individual map: 41.53%. Truth-oracle best of 200: 55.04%; paired difference -0.75 pp $[-2.00, +0.57]$. Pooled votes beat even the best individual on 62 of 97 proteins. Consensus selection: 48.95%; mean-token-likelihood selection: 46.73%. Best of 100 $\rightarrow$ 200 adds 1.20 pp $[0.95, 1.50]$; no claim beyond this budget. Variable sample sizes are scored with whole-map F1, precision and recall.

Errors persist within samples, with some exceptions

Best samples remain poor: 7wz5 9.3%, 8arl 13.1%, 8bux 11.2%, 8dmu 10.5%. 8arl recovers part of an arc; 8dmu still misses extensive distant contacts. 8b6a pooled 66.1% beats its oracle sample 60.2%. 35 of 97 proteins have a better individual sample than pooled votes. Largest gains: 7t9r 29.2% → 52.6%; 7xp9 28.9% → 50.0%. The HTML shows every sample and sorts its gallery by these oracle gains.

Clustering does not yet provide a strong modes test

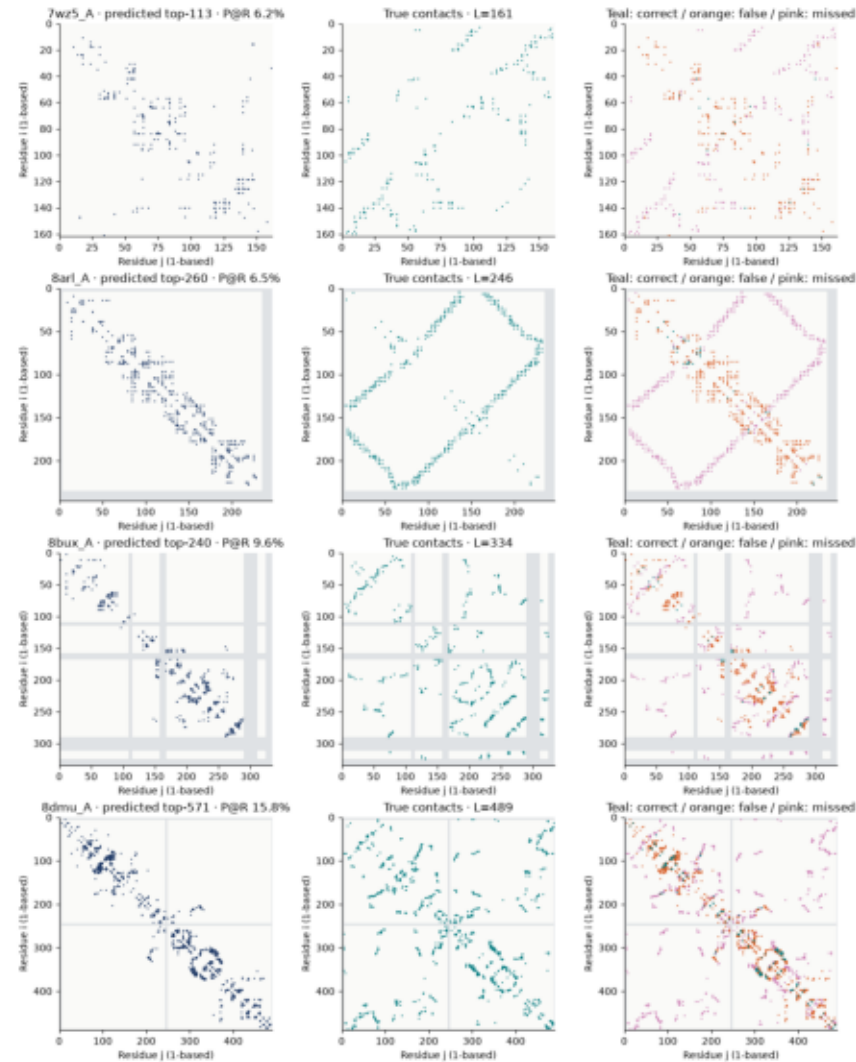
Oracle k=4 cluster voting: 55.73%, tied with pooled 55.79%. Average linkage mostly isolates outliers: median largest training group 96/100. Only 37/194 k=4 fits have at least two eligible test clusters (≥ 10 samples). This null result is limited by the grouping, not proof that useful modes cannot exist. Raw maps, cluster memberships/counts, scores and timings are public.

Validation and interpretation

All 19,400 maps terminate; 582 pooled R-precision checks match the canonical scorer. Independent 100-sample means 55.56% / 55.66% reproduce saved 55.75% within tolerance. Pure inference: 19.15 minutes on one H100; staging/loading recorded separately. Errors within generated maps deserve attention; choosing one sample is insufficient. Contact-map quality does not establish 3D feasibility or a training cause. Next causal probe: true long-range anchors, then score generation of remaining contacts.

cases_1.png

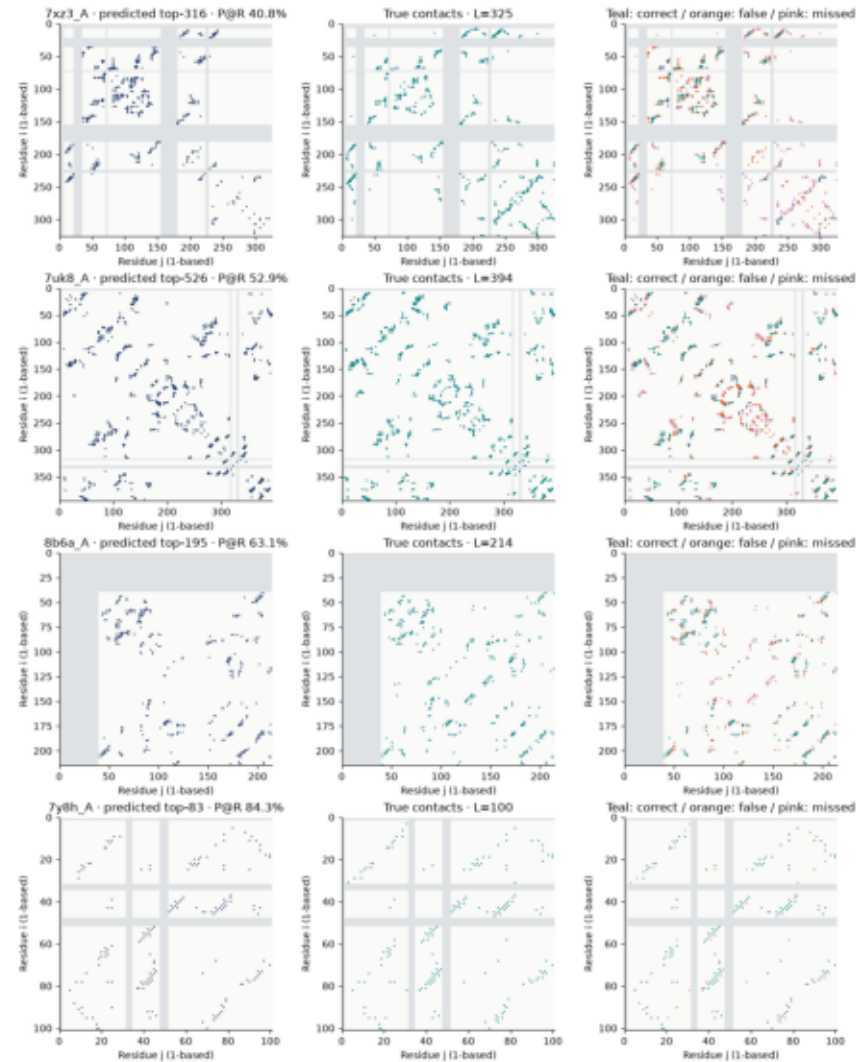
Selected exploratory cases: predicted top-R, truth, and errors. Both halves mirrored; gray bands are unresolved residues.



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$ plot_results.py
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cases_2.png

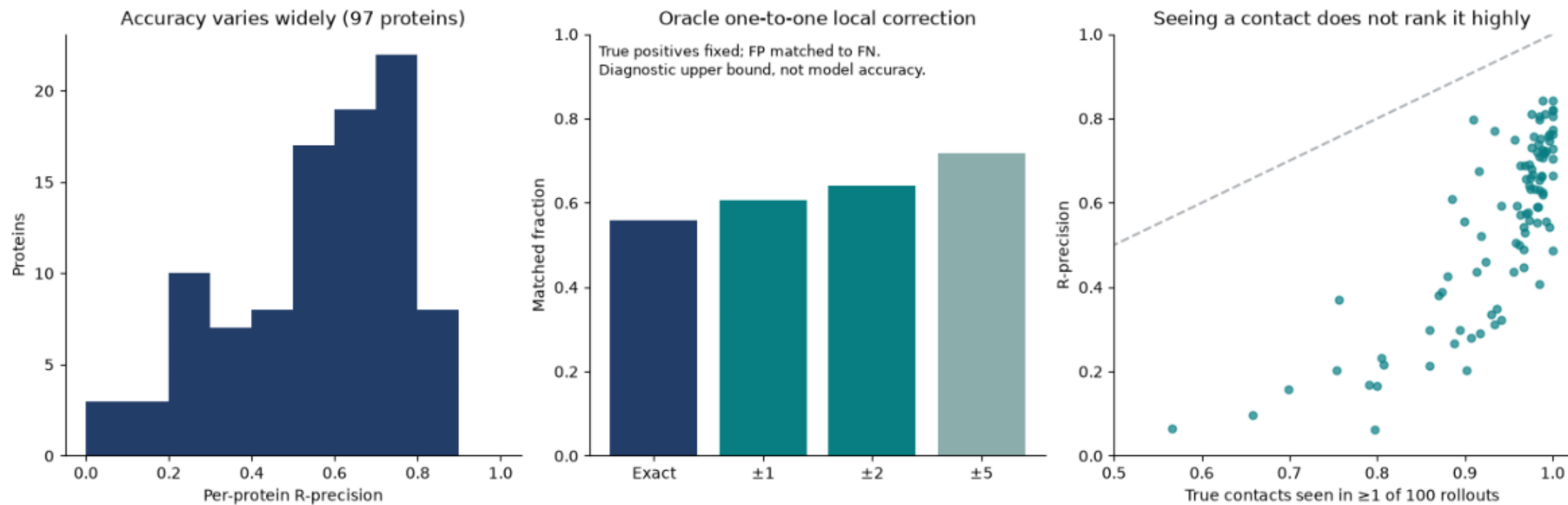
Selected exploratory cases: predicted top-R, truth, and errors. Both halves mirrored; gray bands are unresolved residues.



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$ plot_results.py
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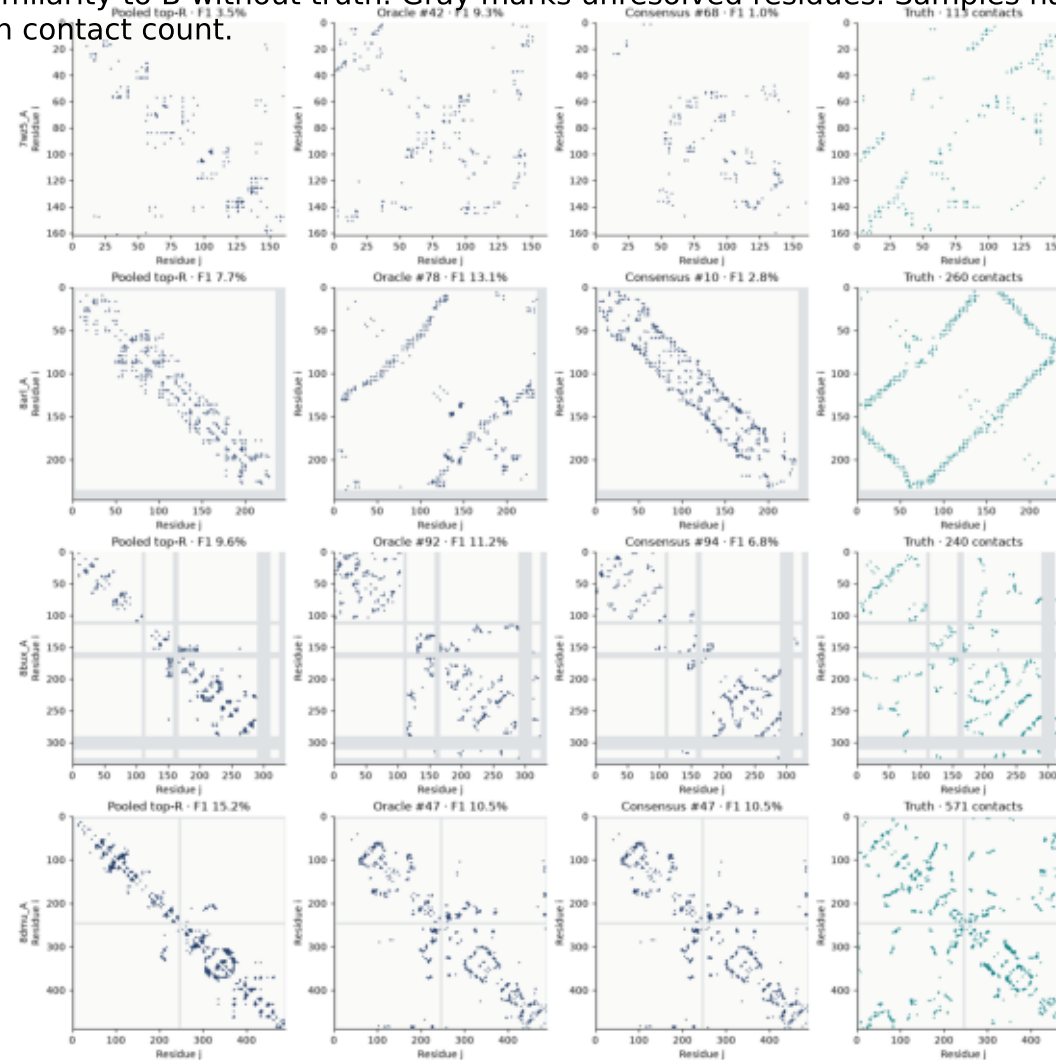
diagnostics.png

All 97 eval-val proteins. Local correction uses a truth-informed maximum one-to-one matching, fixing exact matches first; it is an upper bound, not an improved predictor.



whole_map_cases_1.png

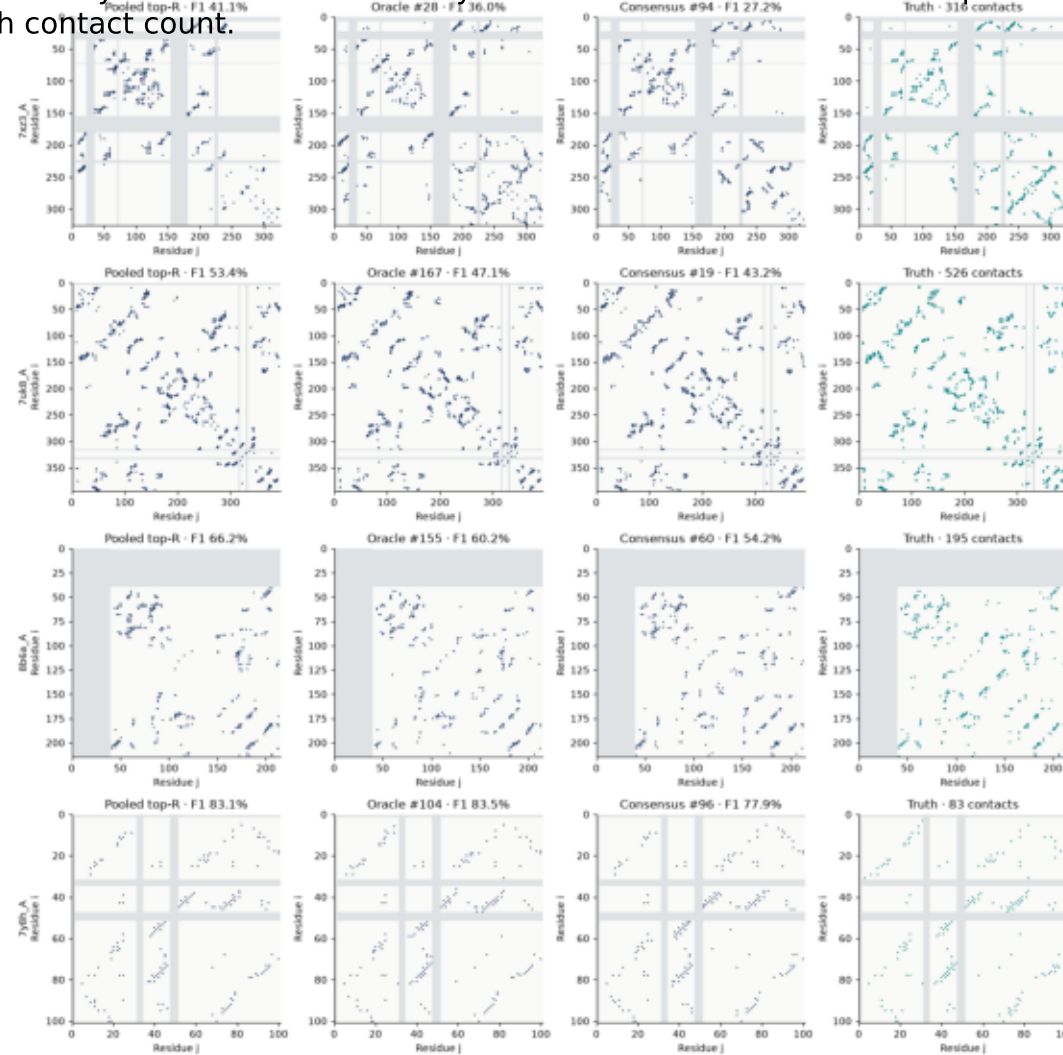
Original eight case studies revisited using fresh samples. The oracle is the individual map maximizing all-range F1 across 200 samples; consensus chooses from A by similarity to B without truth. Gray marks unresolved residues. Samples have variable contact counts; pooled top-R has exactly the truth contact count.



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$ plot_whole_map.py
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whole_map_cases_2.png

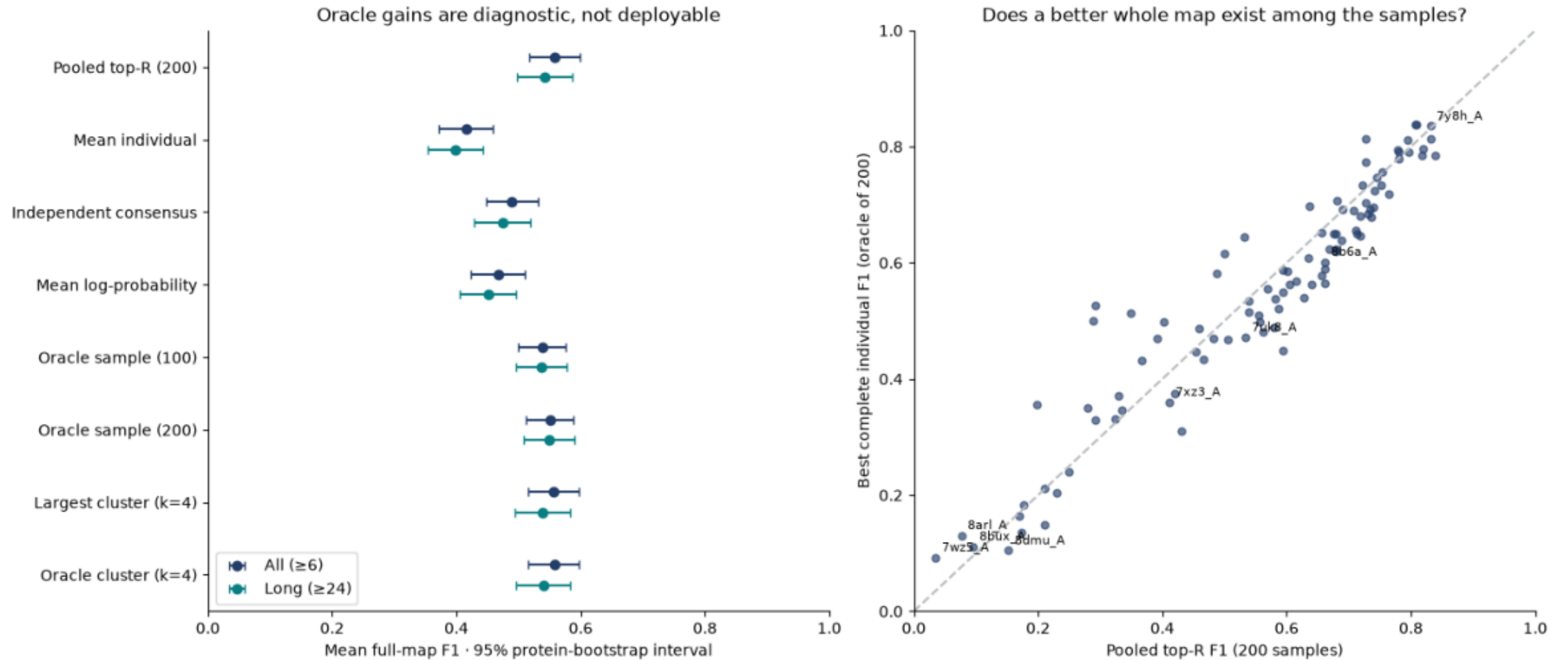
Original eight case studies revisited using fresh samples. The oracle is the individual map maximizing all-range F1 across 200 samples; consensus chooses from A by similarity to B without truth. Gray marks unresolved residues. Samples have variable contact counts; pooled top-R has exactly the truth contact count.



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$ plot_whole_map.py
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whole_map_diagnostics.png

97 eval-val proteins, 200 samples each. Means weight proteins equally. Truth-oracle selectors are upper bounds within these samples; they do not establish physically valid structures. Cluster results use independent-pool assignment and voting, with ≥ 10 assigned samples.



\$ plot_whole_map.py